



Carnegie Mellon University
Master of
Software Engineering

Hexagonal, microkernel

LG Architecture Training Program

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Agenda

Module views

View showing module decomposition

View showing module use

Layered style

MVC style

Data model/domain model

➔ Hexagonal architecture

Microkernel pattern

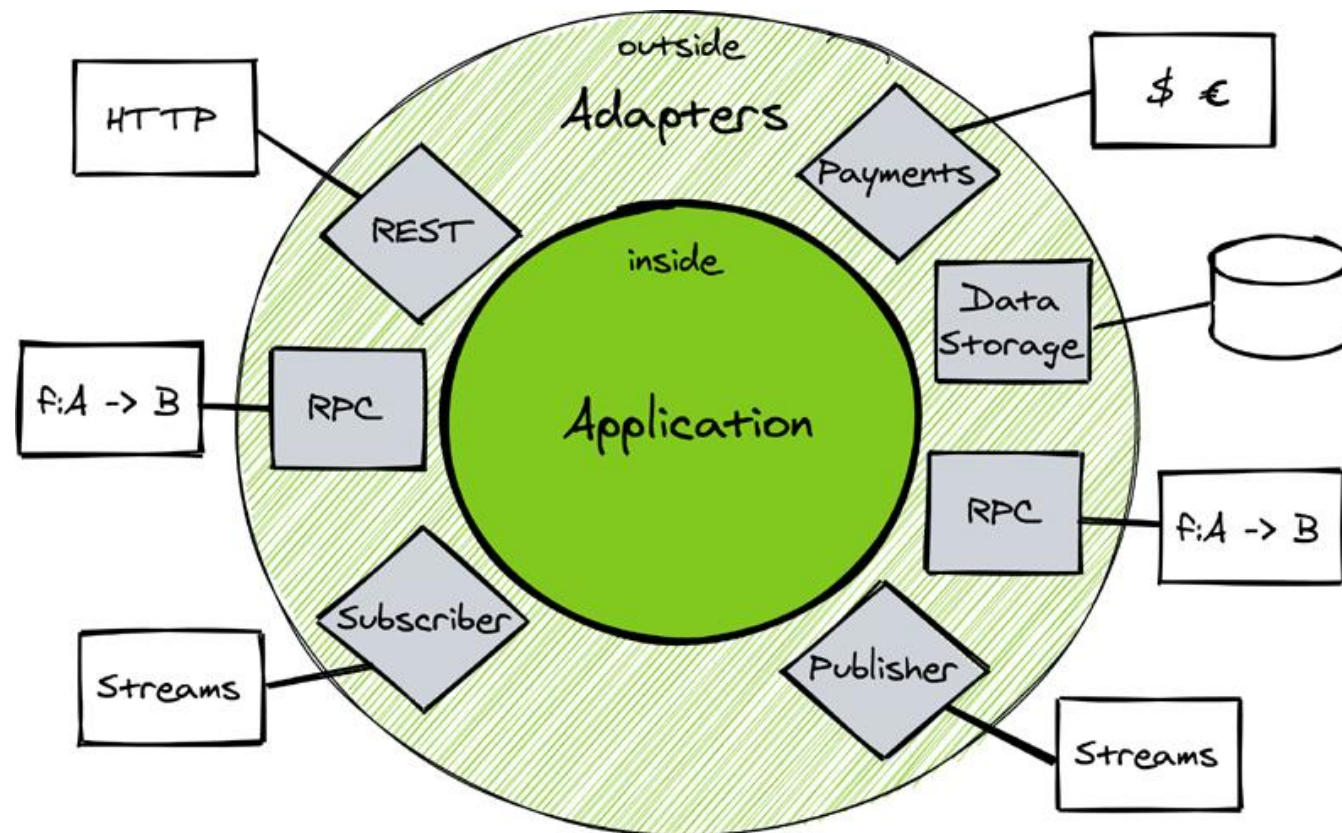
Hexagonal architecture

- Aka Ports and Adapters [Cockburn05][Vernon21]
- Provides a general structure to organize implementation units for applications that interact with multiple GUIs, input devices, databases, and external systems
- Promotes portability and modifiability
- Nice fit for distributed systems, such as microservice-based solutions
- Fits well in the design of applications that follow DDD

Alistair
Cockburn



Main goal: isolate application code from device- or product-specific technologies



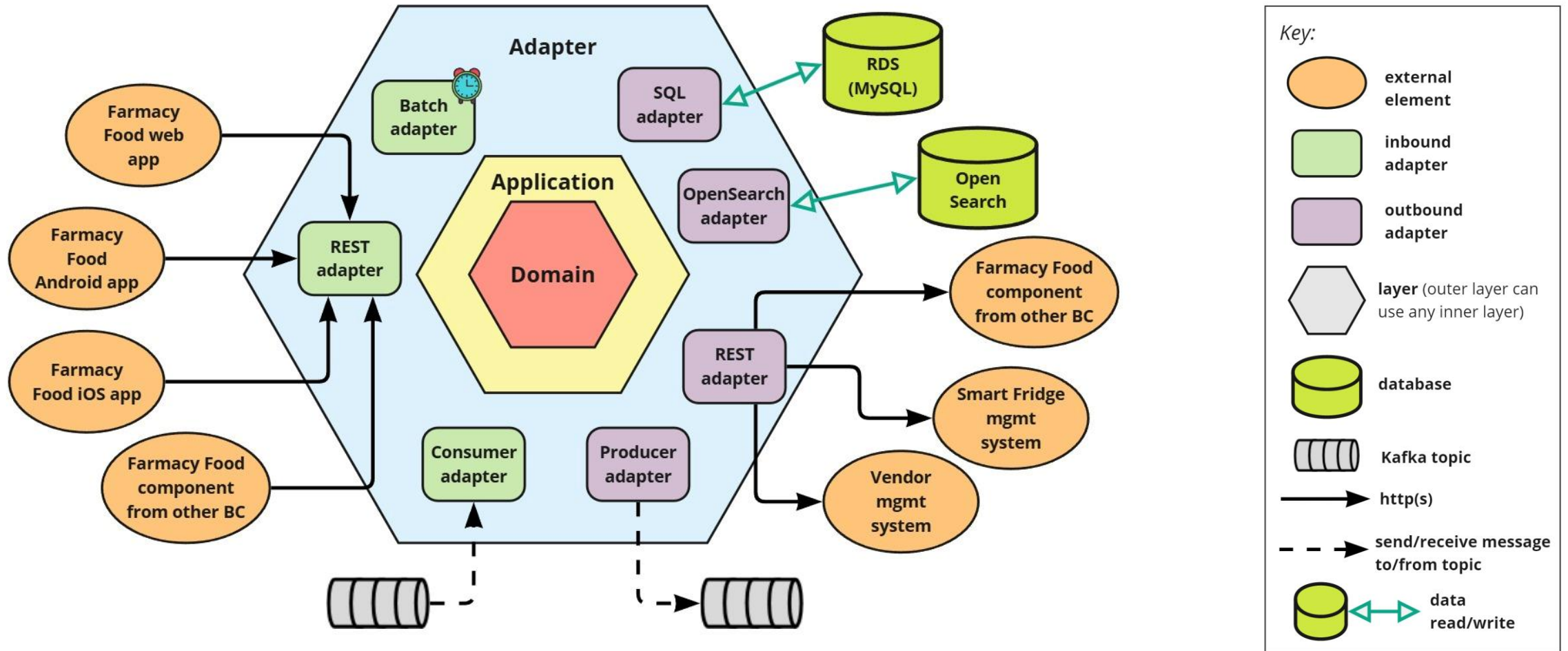
Source: [Vernon21]

This rendition highlights the fact that the hexagon (or the number 6) is not important

Where are the ports?

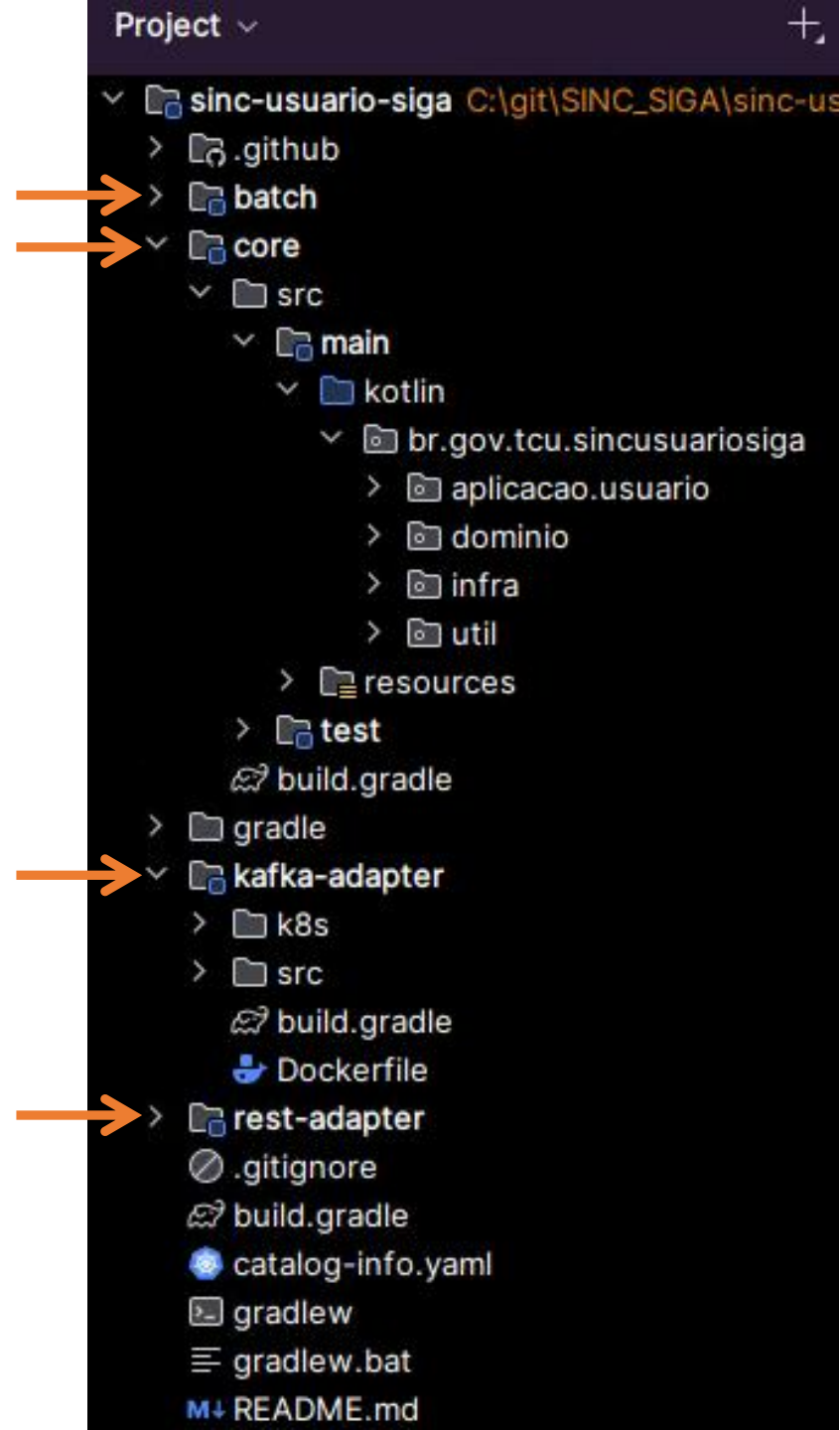


Hexagonal architecture example



Source: github.com/miyagis-forests/farmacy-food-kata

Example of
hexagonal
architecture in the
folder structure



*Adapter modules
depend on core*

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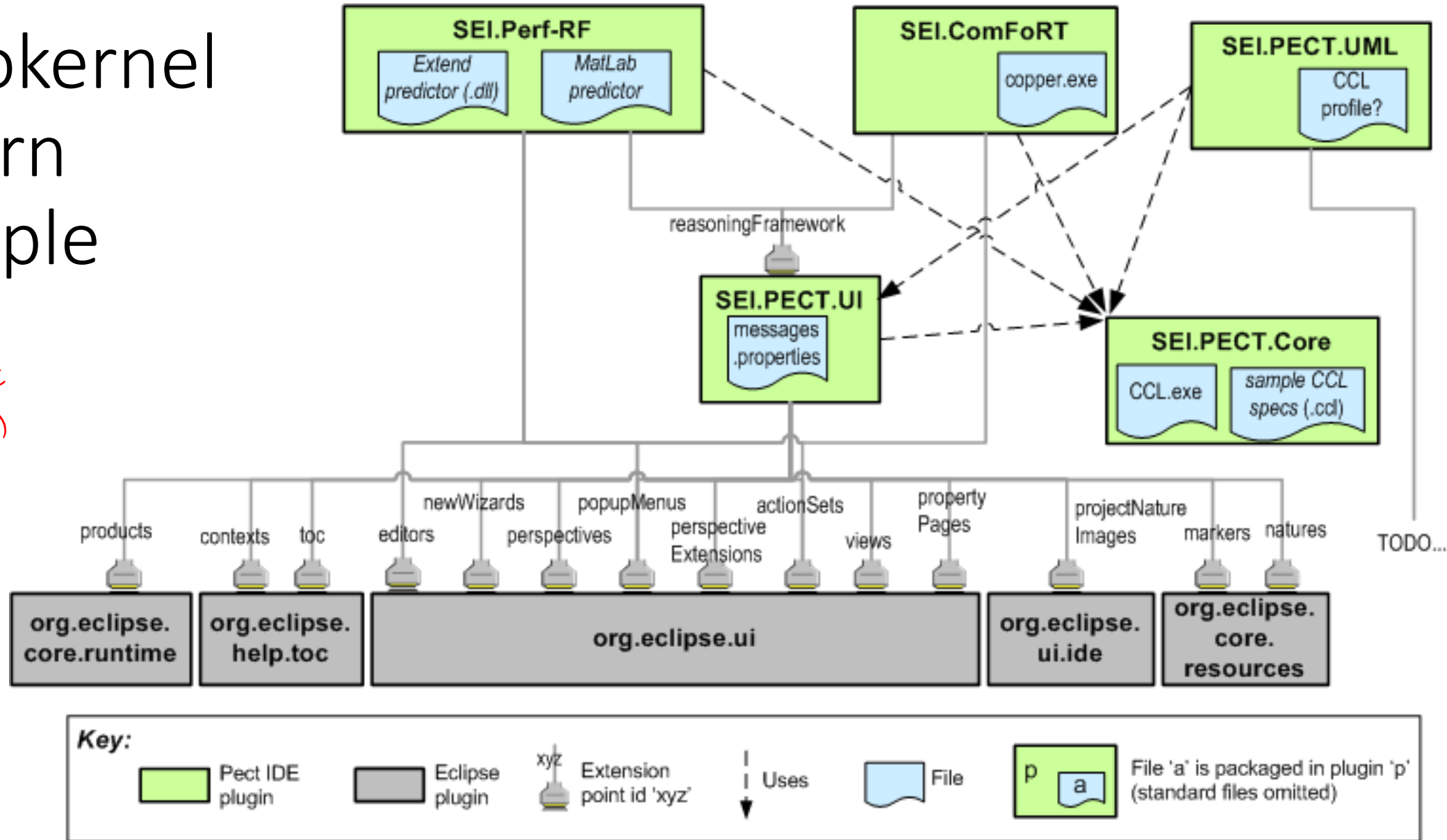
Hexagonal architecture

➔ Microkernel pattern

Microkernel architecture pattern

- A minimal functional **core (microkernel)** provides 1+ extension points
- An extension point defines a contract that is implemented by **plug-in** modules
 - Plug-in binding/registration can be at build-time, initialization-time or even runtime
 - The microkernel can then call the plug-ins that were bound in the system assembly
- A plug-in may itself offer extension points
- Core and plug-ins are created in independent software projects
- Promotes modifiability (extensibility)
- Examples of usage: OS kernels, IDEs, Web browsers, software product lines

Microkernel pattern example



*Can you think of any
LG product that uses
the microkernel
architecture pattern?*





Questions? now or later

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